

scattering by a central potential

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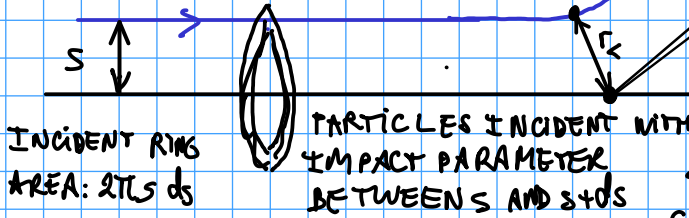
INCIDENT ENERGY $E = \frac{m v_0^2}{2}$

INCIDENT VELOCITY $v_0 = \sqrt{2E/m}$

ANGULAR MOMENTUM $l = S m v_0 = S \sqrt{2mE}$

$S = \frac{l}{\sqrt{2mE}}$

APSIDAL POINT



SCATTERED RING

SOLID ANGLE: $d\Omega = 2\pi \sin\theta d\theta$

$S(\theta, E)$

azimuthal angle ϕ irrelevant by symmetry

PARTICLES EMERGING WITH SCATTERING ANGLE BETWEEN θ AND $\theta+d\theta$

$S \leftrightarrow \theta$

DIFFERENTIAL CROSS SECTION

$\sigma(\theta) d\Omega = \frac{\text{\#PART SCATTERED INTO } d\Omega \text{ PER UNIT TIME}}{\text{INCIDENT INTENSITY}}$

INCIDENT INTENSITY

$I = \frac{\text{\# PART INCIDENT PER UNIT TIME}}{\text{INCIDENT BEAM CROSS SECTION AREA}}$

$I (2\pi S) |ds| = \text{\# part going through ring between } S \text{ and } S+ds = \text{\# part emerging with angle between } \theta \text{ and } \theta+d\theta = I \sigma(\theta) 2\pi \sin\theta |d\theta|$

THEREFORE:

$\sigma(\theta) = \frac{S}{\sin\theta} \left| \frac{ds}{d\theta} \right|$

DEFINE ALSO THE TOTAL CROSS SECTION:

$\sigma_T = \int d\Omega \sigma(\theta)$

REPULSIVE CASE:

$\pi = \theta + 2\psi$
 $\theta = \pi - 2\psi$
 $\psi = \frac{\pi - \theta}{2}$

IN ALL CASES:

$\theta = |\pi - 2\psi|$

ATTRACTIVE CASE:

$\pi = 2\psi - \theta$
 $\theta = 2\psi - \pi$
 $\psi = \frac{\pi + \theta}{2}$

COMPUTING $S(\theta, E)$ (UNIVERSAL METHOD)

$E = \frac{m}{2} \dot{r}^2 + V(r) + \frac{l^2}{2mr^2}$ $\dot{r} = \left[\frac{2}{m} \left(E - V(r) - \frac{l^2}{2mr^2} \right) \right]^{1/2}$ $\dot{r} = \frac{dr}{dt} \Rightarrow dt = \frac{dr}{\dot{r}}$

$l = mr^2 \dot{\theta} \Rightarrow \frac{d\theta}{dt} = \dot{\theta} = \frac{l}{mr^2} \Rightarrow d\theta = \dot{\theta} dt = \frac{l}{mr^2} dt = \frac{l}{mr^2} \frac{dr}{\dot{r}} = \frac{l}{m} \frac{\sqrt{m}}{\sqrt{2E}} \frac{r^2 dr}{\left[1 - \frac{V(r)}{E} - \frac{l^2}{2mE r^2} \right]^{1/2}}$
 $= \frac{l}{\sqrt{2mE}} = S$

$\psi = \int_0^\psi d\psi = \int_{\theta_m}^\pi d\theta = \int_{r_m}^\infty \frac{S r^{-2} dr}{\left[1 - V(r) - \frac{S^2}{r^2} \right]^{1/2}}$
 $u = 1/r \quad du = -r^{-2} dr \quad u_m = 1/r_m$

$\theta(S) = \left| \pi - 2 \int_0^{u_m} \frac{S du}{\left[1 - \frac{V(u)}{E} - S^2 u^2 \right]^{1/2}} \right|$

From here, compute $\left| \frac{ds}{d\theta} \right| = \frac{1}{\left| \frac{d\theta}{ds} \right|}$ and use it to get $\sigma(\theta) = \frac{S}{\sin\theta} \left| \frac{ds}{d\theta} \right|$

UP TO HERE, RESULTS ARE VALID FOR ANY CENTRAL POTENTIAL $V(r)$

Coulomb Scattering

FORCE $F = \frac{Z Z' e^2}{r^2}$

$f = -\frac{k}{r^2}$ $V = -\frac{k}{r}$
 $k = -Z Z' e^2$

$E > 0$	hyperbolic
$E = 0$	parabolic
$E < 0$	elliptic

ONLY FOR $V = -\frac{k}{r}$!!!

Scattering: $E > 0$

$\frac{1}{r} = G [1 + E \cos(\tilde{\theta} - \tilde{\theta}')] \quad G = \frac{mk}{l^2} = -\frac{m}{l^2} Z Z' e^2$

Eccentricity $E = \left[1 + \frac{2E l^2}{m k^2} \right]^{\frac{1}{2}} = \left[1 + \frac{2E l^2}{m (Z Z' e^2)^2} \right]^{\frac{1}{2}} = \left[1 + \left(\frac{2sE}{Z Z' e^2} \right)^2 \right]^{\frac{1}{2}} > 1$

$l = s m \tilde{v}_0 = s \sqrt{2mE} \rightarrow \frac{2l^2 E}{m} = \frac{s^2 Z^2 e^4 E^2}{m} = (2sE)^2$

2 SITUATIONS: i) REPULSIVE CASE $Z Z' > 0 \Rightarrow k = -Z Z' e^2 < 0$

ii) ATTRACTIVE CASE $Z Z' < 0 \Rightarrow k = -Z Z' e^2 > 0$ (ALSO GRAVITY $k = G m m_2$)

i) REPULSIVE CASE: $k < 0 \quad G = \frac{mk}{l^2} < 0$

$\frac{1}{r} = G [1 + E \cos(\tilde{\theta} - \tilde{\theta}')] > 0 \Leftrightarrow 1 + E \cos(\tilde{\theta} - \tilde{\theta}') < 0 \Leftrightarrow E \cos(\tilde{\theta} - \tilde{\theta}') < -1 \Leftrightarrow \cos(\tilde{\theta} - \tilde{\theta}') < -\frac{1}{E}$

We need the cosine to be negative and larger in magnitude than $1/E$

In this case, $\tilde{\theta} = \tilde{\theta}'$ is not a physical point, because $\cos(\tilde{\theta} - \tilde{\theta}') = \cos(0) = 1 \Rightarrow \frac{1}{r} = G[1+E] < 0$ UNPHYSICAL

SO $\tilde{\theta} = \tilde{\theta}'$ IS IN THE MIDDLE OF THE INACCESSIBLE REGION OF THE HYPERBOLA, AND $\tilde{\theta} = \tilde{\theta}' \pm \pi$ WILL BE THE PERIAPSIS, SO WE CHOOSE $\tilde{\theta}' = \pi$ AND WE GET

$\cos(\tilde{\theta} - \tilde{\theta}') = \cos(\tilde{\theta} - \pi) = \cos \tilde{\theta} \cos \pi + \sin \tilde{\theta} \sin \pi = -\cos \tilde{\theta}$

$\frac{1}{r} = \left(-\frac{mk}{l^2} \right) [-1 - E \cos(\tilde{\theta} - \tilde{\theta}')] = \frac{m Z Z' e^2}{l^2} [E \cos \tilde{\theta} - 1]$

MAXIMUM AT $\tilde{\theta} = 0 \Rightarrow$

MINIMUM r : $\frac{1}{r_m} = \frac{m Z Z' e^2}{l^2} [E - 1]$

SO THE ANGLE AT THE APSIDAL POINT IS $\tilde{\theta} = 0$

WE NOW NEED $\psi = \tilde{\theta}(r \rightarrow \infty) = \tilde{\theta}(\frac{1}{r} \rightarrow 0)$

[IN THE REPULSIVE CASE]
 $\theta = \pi - 2\psi \Rightarrow \psi = \frac{\pi - \theta}{2}$

$0 = E \cos \psi - 1 \Leftrightarrow \frac{1}{E} = \cos \psi = \cos\left(\frac{\pi - \theta}{2}\right) = \cos \frac{\pi}{2} \cos \frac{\theta}{2} + \sin \frac{\pi}{2} \sin \frac{\theta}{2} = \sin \frac{\theta}{2}$

ii) ATTRACTIVE CASE: $k > 0 \quad G = \frac{km}{l^2} > 0$

TAKE $\tilde{\theta}' = 0 \quad \tilde{\theta} = \tilde{\theta}' = 0 \Rightarrow \frac{1}{r} = G [1 + E \cos(0)] = G [1 + E]$

APSIDAL POINT IS ALSO AT $\tilde{\theta} = 0$

WE NOW NEED $\psi = \tilde{\theta}(r \rightarrow \infty) = \tilde{\theta}(\frac{1}{r} \rightarrow 0)$

MAX AT $\tilde{\theta} = 0$

MINIMUM r : $\frac{1}{r_m} = G [1 + E]$

[IN THE ATTRACTIVE CASE]
 $\theta = 2\psi - \pi \Rightarrow \psi = \frac{\pi + \theta}{2}$

$0 = 1 + E \cos \psi \Leftrightarrow -\frac{1}{E} = \cos \psi = \cos\left(\frac{\pi + \theta}{2}\right) = \cos \frac{\pi}{2} \cos \frac{\theta}{2} - \sin \frac{\pi}{2} \sin \frac{\theta}{2} = -\sin \frac{\theta}{2}$

We now combine both cases:

$$\cot^2 \frac{\theta}{2} = \frac{1}{\tan^2 \frac{\theta}{2}} = \frac{\cos^2 \frac{\theta}{2}}{\sin^2 \frac{\theta}{2}} = \frac{1 - \sin^2 \frac{\theta}{2}}{\sin^2 \frac{\theta}{2}} = \frac{1}{\sin^2 \frac{\theta}{2}} - 1 = \epsilon^2 - 1 = \left(\frac{2sE}{zz'e^2} \right)^2 \Rightarrow \boxed{s = s(\theta, \epsilon) = \left| \frac{zz'e^2}{2E} \right| \cot \frac{\theta}{2}}$$

$$\frac{ds}{d\theta} = \left| \frac{zz'e^2}{2E} \right| \frac{d}{d\theta} \left(\frac{\cos \frac{\theta}{2}}{\sin \frac{\theta}{2}} \right) = \left| \frac{zz'e^2}{2E} \right| \left(\frac{-\sin \frac{\theta}{2}}{2 \sin^2 \frac{\theta}{2}} - \frac{1}{2} \frac{\cos \frac{\theta}{2}}{\sin^2 \frac{\theta}{2}} \right) = - \left| \frac{zz'e^2}{4E} \right| \frac{1}{\sin^2 \frac{\theta}{2}}$$

$$\sigma(\theta) = \frac{s}{\sin \theta} \left| \frac{ds}{d\theta} \right| = \frac{1}{2} \left(\frac{zz'e^2}{2E} \right)^2 \frac{1}{\sin \frac{\theta}{2} \cos \frac{\theta}{2}} \frac{1}{\sin^2 \frac{\theta}{2}}$$

$$\boxed{\sigma(\theta) = \frac{1}{4} \left(\frac{zz'e^2}{2E} \right)^2 \csc^4 \left(\frac{\theta}{2} \right)}$$

Rutherford scattering cross section
(for Coulomb force)

Remarks:

- 1) The cross section is the same for the repulsive and attractive cases, since it is proportional to $(zz')^2$.
- 2) It diverges like E^{-2} at low energies.
- 3) It diverges like θ^{-4} (VERY STRONG!!!) for small angles θ .
- 4) This classical result is exactly the same as the result obtained in Quantum Mechanics.

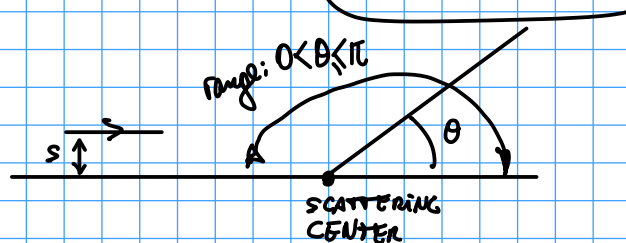
Total scattering cross section:

$$\sigma_T = \int \sigma(\Omega) d\Omega$$

In this case $\sigma(\theta) = \text{const} \frac{1}{\sin^4(\frac{\theta}{2})}$

$$d\Omega = 2\pi \sin \theta d\theta = 4\pi \sin \frac{\theta}{2} \cos \frac{\theta}{2} d\theta$$

$$\sigma_T = \frac{4\pi}{4} \left(\frac{zz'e^2}{2E} \right)^2 \int_0^\pi \frac{\sin \frac{\theta}{2} \cos \frac{\theta}{2}}{\sin^4 \frac{\theta}{2}} d\theta$$



NOTATION:

Goldstein

Others

$$\sigma(\Omega) \leftrightarrow \frac{d\sigma}{d\Omega} \text{ DIFFERENTIAL SCATTERING CROSS SECTION}$$

$$\sigma_T \leftrightarrow \sigma \text{ TOTAL SCATTERING CROSS SECTION}$$

$$\sim \int_0^\pi \frac{d\theta}{(\theta/2)^3} = 8 \int_0^\pi \frac{d\theta}{\theta^3} = 8 \left. \frac{\theta^{-2}}{(-2)} \right|_0^\pi = +\infty \text{ STRONGLY DIVERGENT}$$

CLASSICAL

$$\sigma_T = +\infty \text{ for any long range potential}$$

QUANTUM

$$\sigma_T = +\infty \text{ for some long range potentials } V(r) \text{ do not decay fast enough at } r \rightarrow \infty$$

CAVEAT

1) IN A DENSE CONDUCTING MATERIAL:

$$V = -\frac{k}{r} \longrightarrow V(r) = -\frac{k}{r} e^{-r/\lambda}$$

SCREENED COULOMB POTENTIAL ("YUKAWA")

(In non-conducting materials $V(r)$ may take a different form, but still will vanish much faster than $1/r$)

Chapter 8 Hamiltonian and Hamilton's Eqs of Motion

MIDTERM REMARK

In polar coordinates

$$\frac{\partial L}{\partial \dot{\theta}} = p_{\theta} = l_z$$

$$L = L(r, \dot{r}, p_{\theta})$$

Lagrange Eqs

$$L = T - V = L(q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n, t)$$

$$0 = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} \quad p_i = \frac{\partial L}{\partial \dot{q}_i}$$

Legendre Transformation $q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n \rightarrow q_1, \dots, q_n, p_1, \dots, p_n$

$$f(x, y) \Rightarrow df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$$

In Thermodynamics

Fluid at constant N

$$1^{st} \text{ Law } dU = dQ + dW = TdS - p dV$$

$$U = U(S, V) \quad \text{Internal Energy}$$

We want T, V as variables

$F = A = U - TS$ Helmholtz Free Energy

$$dA = dU - d(TS) = TdS - p dV - TdS - S dT = -S dT - p dV \Rightarrow A = A(T, V)$$

In general

$$f(x, y) = u dx + v dy \quad u = \frac{\partial f}{\partial x} \quad v = \frac{\partial f}{\partial y} \quad (x, y) \rightarrow (u, y)$$

$$g = f - ux \Rightarrow dg = df - d(ux) = u dx + v dy - u dx - x du = -x du + v dy$$

$$g = g(u, y)$$

$$\text{let's look at } L(q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n) \quad \left(p_i = \frac{\partial L}{\partial \dot{q}_i} \right)$$

$$H = - \left(L - \sum_i p_i \dot{q}_i \right) \quad dH = -dL + \sum_i d(p_i \dot{q}_i) = - \sum_i \left(\frac{\partial L}{\partial q_i} dq_i + \frac{\partial L}{\partial \dot{q}_i} d\dot{q}_i \right) + \sum_i (p_i d\dot{q}_i + \dot{q}_i dp_i)$$

$$dH = - \frac{\partial L}{\partial t} dt + \sum_i (\dot{q}_i dp_i - \frac{\partial L}{\partial q_i} dq_i) \Rightarrow H = H(p_1, \dots, p_n, q_1, \dots, q_n, t)$$

$$\frac{\partial H}{\partial p_i} = \dot{q}_i \quad \frac{\partial H}{\partial q_i} = - \frac{\partial L}{\partial q_i} = - \dot{p}_i \quad \frac{\partial H}{\partial t} = - \frac{\partial L}{\partial t}$$

CANONICAL EQS OF MOTION